Problem definition

Choose n nodes form m nodes to from a bicycle path without duplilcated nodes ,by maximizing the comfort and minimizing the distance.

Distance Matrix:

- A symmetric matrix filled with random distances

The four comfort Matrix:

- A symmetric matrix with scores from 1 to 5

For safety matrix and slope matrix, transform it to make it the higher the better(6 - x)

Example Data

Multi-Objective Mathematical Problem:

min -> distance

max -> comfort

$1 \le aij \le 5$,beauty

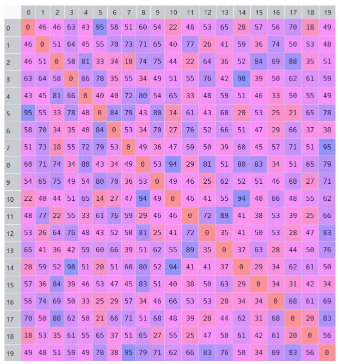
$1 \le bij \le 5$, roughness

$1 \le sij \le 5$, safety

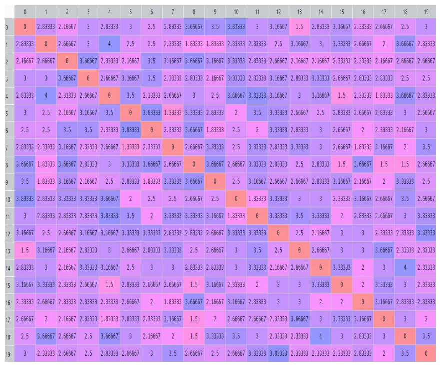
$1 \le lij \le 5$, slope

$10 \le dij \le 100$, distance

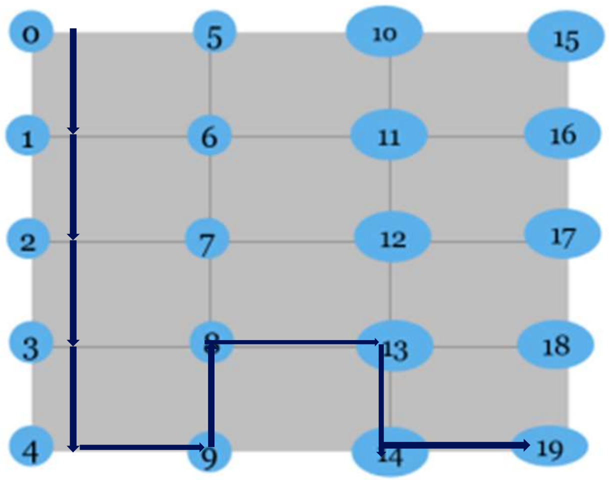
$\frac{a+2b+s+2l}{6} \ge 20$,comfort



Distance Matrix



Comfort Matrix



**we multiply the parameters roughness and slope with 2, since we believe that they are the most important.

Yudie Zheng , Myrto Pieridou ,Christian Steenis

Genetic Algorithm

Initialization

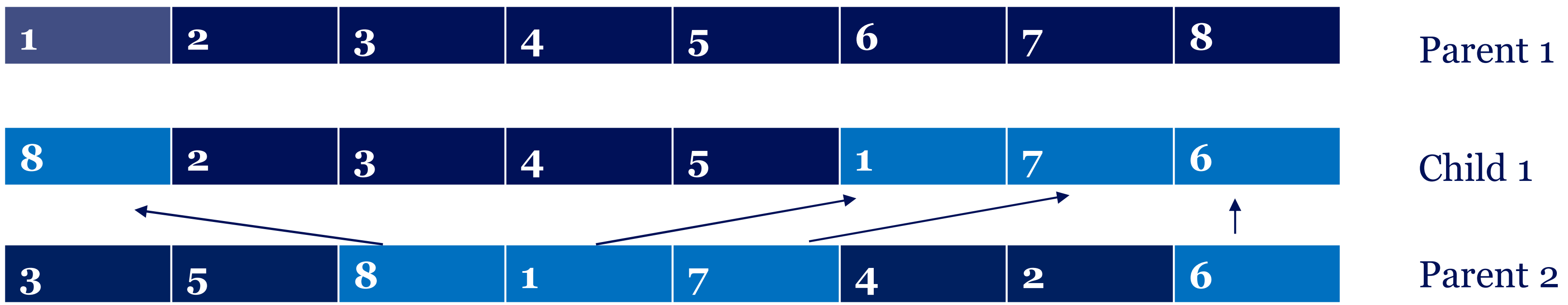
- Initialize random routes

Fitness Function

- Minimize the total distance while maximize the overall comfort

Generic Operations

- Crossover

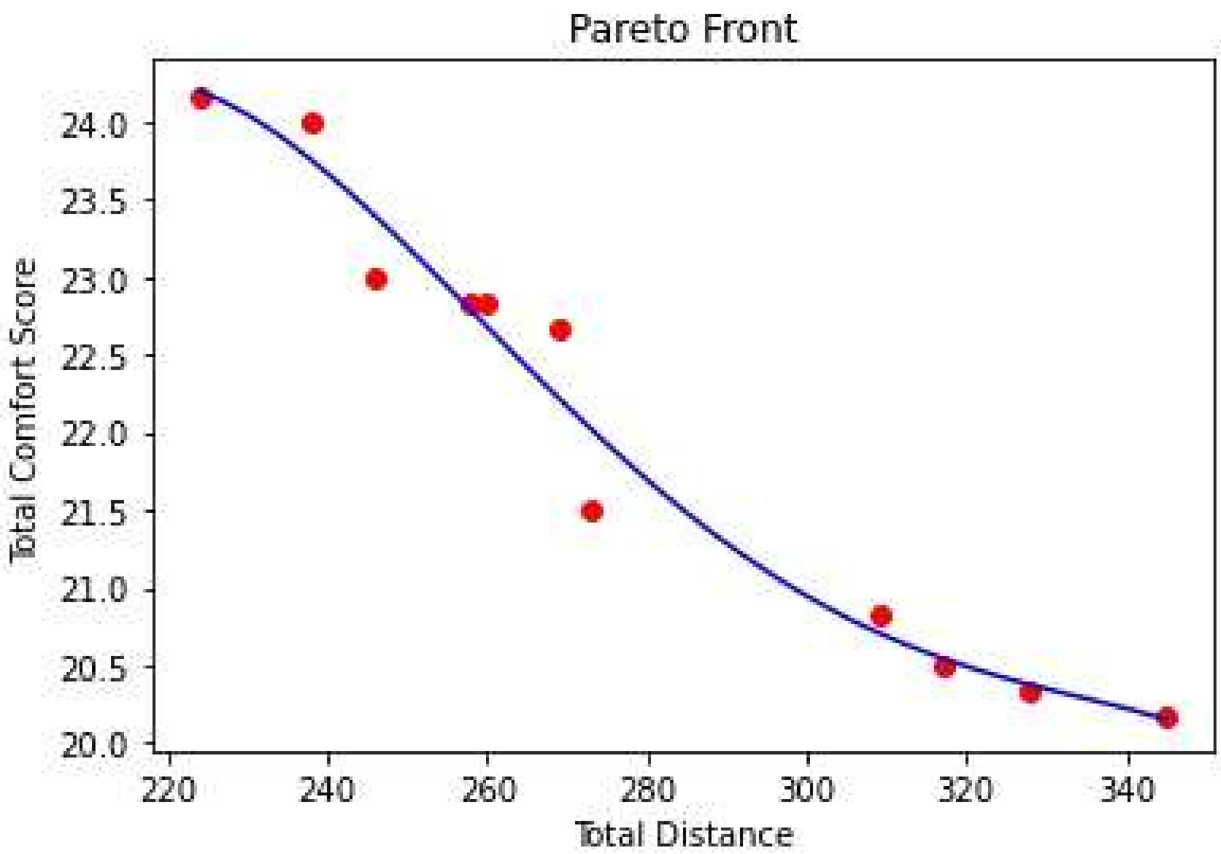


- Mutation
  - Randomly shuffle the data

Results

The red dots are non-dominated solutions, meaning no other solutions are superior in all objectives simultaneously.

When total distance increases, the total comfort score and vise verse.



Discussion and Summary

In this study, genetic algorithm was used to solve a multi-objective optimization problem focusing on bicycle route planning.

The objectives were to minimize the total distance traveled and maximize the comfort of the route, considering scenery, smoothness, safety, and slope.

The resulting pareto front shows the trade-off between the total distance and total comfort score.

The genetic algorithm maintains a genetic diversity that helps avoid premature convergence to local optimal solutions.